Scalability & Future Plan

Date	19 October 2026
Team ID	
Project Name	Health360: Clinical and healthcare risk analysis
Maximum Marks	1 Mark

Scalability & Future Plan:

Health360 is a clinical and healthcare risk analysis solution intended to help healthcare teams review patient-related risk indicators, identify cases that may need attention, and support timely, evidence-informed decisions. This plan describes how the solution can grow safely as users, clinical records, and analytical workloads increase. It is a proposed roadmap; clinical outputs should support, not replace, qualified healthcare professionals.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited capacity for concurrent users and large patient-record batches.	Slower analysis and reduced usability during peak clinical hours.	**High**
2	Data may come from different formats or systems without a consistent integration layer.	Duplicate, incomplete, or inconsistent records can affect analysis quality.	**High**
3	Security, audit logging, and model monitoring may need strengthening as adoption grows.	Higher privacy/compliance risk and less visibility into unexpected results.	**High**

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for a limited number of users and workflows.	Use a stateless application layer, load balancing, and horizontal scaling; apply role-based access for clinical teams.
2	Data Storage	Records and inputs may require more consistent structuring.	Adopt a validated data model, encrypted database and object storage, retention rules, backups, and controlled data imports.
3	Performance	Large datasets and risk calculations may increase response time.	Add asynchronous job queues, caching where appropriate, database indexing, pagination, and performance monitoring.
4	Security	Controls need to scale with sensitive healthcare data.	Enforce least-privilege access, encryption in transit and at rest, audit trails, secure secrets, regular reviews, and incident-response procedures.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Core data validation, role-based access, audit logs, and baseline performance tests.	0–3 months	Safer data handling and a measurable baseline for reliability.
Phase 2	Scalable deployment, background processing, backups, and monitoring dashboards.	3–6 months	Improved responsiveness and service continuity as usage grows.
Phase 3	Integrations with approved healthcare data sources using standardized interfaces.	6–9 months	Less manual entry and more consistent data exchange.
Phase 4	Model validation, drift/bias monitoring, explainability, and clinician feedback workflow.	9–12 months	More transparent and monitored risk outputs with human oversight.

Success measures: Track response time, successful processing rate, system availability, data-quality errors, security events, and clinician feedback. Validate analytical performance on appropriate data before deployment and review it regularly. Any clinical use must follow applicable privacy, security, and healthcare requirements.